Random Variable (R.V.)

A **Random Variable (X)** is a mathematical concept that links the results of a random experiment to numerical values. It is a **function** that maps every outcome in the **sample space (S)** to a unique real number.

- **Domain:** The **Sample Space (S)** (The set of all possible outcomes of the experiment).
- **Range:** A **subset of Real Numbers ($\mathbb{R}$)** (The possible numerical values the random variable can take).

Example 1: Tossing Two Coins

Experiment: Tossing two coins.
Random Variable (X): "Number of Heads".

1. **Determine the Sample Space (S):**
   The possible outcomes when tossing two coins are:

$$S = \{HH, HT, TH, TT\}$$

2. **Determine the Values of the Random Variable (X):**
   The function X assigns a numerical value (the count of Heads) to each sample point:
   - For the outcome HH: $X(HH) = \mathbf{2}$ (two Heads)
   - For the outcome HT: $X(HT) = \mathbf{1}$ (one Head)

- For the outcome TH: $X(TH) = \mathbf{1}$ (one Head)
- For the outcome TT: $X(TT) = \mathbf{0}$ (zero Heads)

The values that the Random Variable X can take are $\{0, 1, 2\}$.

Example 2: Tossing Three Coins

Experiment: Tossing three coins.
Random Variable (X): "Number of Heads".

1. **Determine the Sample Space (S):**
   The possible outcomes when tossing three coins are (Total $2^3 = 8$ outcomes):

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

2. **Determine the Sample Points and Corresponding Values for X:**
   The function X assigns the count of Heads to each sample point:

Sample Point (ω)	Description	Value of R.V. ($X(\omega)$)
HHH	Three Heads	$\mathbf{3}$
HHT	Two Heads, One Tail	$\mathbf{2}$
HTH	Two Heads, One Tail	$\mathbf{2}$
THH	Two Heads, One Tail	$\mathbf{2}$
HTT	One Head, Two Tails	$\mathbf{1}$
THT	One Head, Two Tails	$\mathbf{1}$
TTH	One Head, Two Tails	$\mathbf{1}$
TTT	Zero Heads	$\mathbf{0}$

The values that the Random Variable X can take are $\{0, 1, 2, 3\}$.

2. Types of Random Variables

A Random Variable (R.V.) is classified based on the nature of the numerical values it can assume.

- **Discrete Random Variable:** This is a variable whose possible values are countable. The values are typically integers and have gaps between them (isolated values).
  - *Examples:* The number of defective items in a batch, the number of emails received per hour, or the number of heads in coin tosses (as in our examples).
- **Continuous Random Variable:** This is a variable that can take on any value within a given range or interval. The number of possible values is uncountable.
  - *Examples:* The height of a person, the temperature of a room, or the time taken to complete a task.

3. Discrete Random Variable Tools

The behavior of a Discrete Random Variable (X) is fully described by its probability distribution, which is managed using the following tools:

Probability Mass Function (PMF)

The **Probability Mass Function ($P(X = x)$ or $p(x)$)** is a function that provides the probability for each specific value x that the discrete random variable X can take.

- **Conditions for a function to be a PMF:**
  1. **Non-Negativity:** The probability for any given value x_i must be non-negative:

     $$P(x_i) \geq 0 \quad \text{for all } i$$

  2. **Total Probability:** The sum of the probabilities over all possible values of the random variable must equal 1:

     $$\sum P(x_i) = 1$$

Probability Distribution

The **Probability Distribution** is a table or list that systematically pairs every possible value of the random variable (x_i) with its corresponding probability ($p_i = P(X = x_i)$).

Mean (Expected Value)

The **Mean (μ)**, or **Expected Value ($E(X)$)**, represents the long-run average value of the random variable if the experiment were repeated many times. It is the center of the probability distribution.

$$\mu = E(X) = \sum x_i p_i$$

Variance

The **Variance (σ^2)** measures the spread or dispersion of the probability distribution around its mean. A higher variance means the values are more spread out from the expected value.

$$\sigma^2 = \sum p_i(x_i - \mu)^2$$

The **Standard Deviation** (σ) is the square root of the variance and is often preferred because it is in the same units as the random variable X.

Application: Two Coin Tossed Example

Let's use the experiment of tossing two coins to illustrate these tools.
Random Variable (X): The number of Heads.

1. Identify Values and Calculate PMF

- **Sample Space:** $S = \{HH, HT, TH, TT\}$, so $n(S) = 4$.
- **Possible Values for X (x_i):** $\{0, 1, 2\}$.

We calculate the probability $P(X = x)$ for each value:

Value of X (x_i)	Outcome(s)	Probability $P(X = x_i) = p_i$
0	TT	$P(X = 0) = \frac{1}{4}$
1	HT, TH	$P(X = 1) = \frac{2}{4} = \frac{1}{2}$
2	HH	$P(X = 2) = \frac{1}{4}$

2. Construct the Probability Distribution

The table above is the **Probability Distribution** for the random variable X.

Verification of PMF Conditions:

1. **Non-Negativity:** $1/4, 1/2, 1/4$ are all ≥ 0. **(Condition 1 satisfied)**
2. **Total Probability:** $\sum P(x_i) = \frac{1}{4} + \frac{2}{4} + \frac{1}{4} = \frac{4}{4} = 1$. **(Condition 2 satisfied)**

3. Calculate the Mean (Expected Value)

$$\mu = E(X) = \sum x_i p_i$$

x_i	p_i	$x_i p_i$
0	$1/4$	$0 \times 1/4 = 0$
1	$2/4$	$1 \times 2/4 = 2/4$
2	$1/4$	$2 \times 1/4 = 2/4$
Sum	1	$\sum x_i p_i = 4/4 = \mathbf{1}$

The **Mean** or **Expected Value** is $\mu = 1$. (We expect 1 Head on average over many trials).

4. Calculate the Variance

$$\sigma^2 = \sum p_i (x_i - \mu)^2$$

x_i	p_i	$(x_i - \mu)$	$(x_i - \mu)^2$	$p_i (x_i - \mu)^2$
0	$1/4$	$0 - 1 = -1$	$(-1)^2 = 1$	$1/4 \times 1 = 1/4$
1	$2/4$	$1 - 1 = 0$	$(0)^2 = 0$	$2/4 \times 0 = 0$
2	$1/4$	$2 - 1 = 1$	$(1)^2 = 1$	$1/4 \times 1 = 1/4$

x_i	p_i	$(x_i - \mu)$	$(x_i - \mu)^2$	$p_i(x_i - \mu)^2$
Sum	1			$\sum p_i(x_i - \mu)^2 = 2/4 = \mathbf{0.5}$

The **Variance** is $\sigma^2 = 0.5$.

Mean And Variance Of Discrete Random Variables

The **Mean** and **Variance** are descriptive measures that summarize the central tendency and spread of a random variable's probability distribution.

Definition

- **Mean (μ or Expected Value, $E(X)$):** The average value of the random variable over a very large number of trials. It represents the central location of the distribution.
- **Variance (σ^2 or $V(X)$):** The average squared deviation of the random variable's values from its mean. It measures the dispersion or spread of the distribution.

Formula for a Discrete Random Variable (X)

Measure	Formula
Mean (Expected Value)	$\mu = E(X) = \sum x_i P(X = x_i)$

Measure	Formula
Variance	$\sigma^2 = V(X) = \sum (x_i - \mu)^2 P(X = x_i)$
Alternate Variance Formula	$\sigma^2 = E(X^2) - [E(X)]^2$
where	$E(X^2) = \sum x_i^2 P(X = x_i)$

Explanation with Example: Rolling a Fair Die

Experiment: Rolling a single fair six-sided die.
Random Variable (X): The number appearing on the die.

x_i (Value)	$P(X = x_i) = p_i$	$x_i p_i$	x_i^2	$x_i^2 p_i$
1	$1/6$	$1/6$	1	$1/6$
2	$1/6$	$2/6$	4	$4/6$
3	$1/6$	$3/6$	9	$9/6$
4	$1/6$	$4/6$	16	$16/6$
5	$1/6$	$5/6$	25	$25/6$
6	$1/6$	$6/6$	36	$36/6$
Sum	$6/6 = 1$	$21/6$		$91/6$

- **1. Calculate the Mean ($E(X)$):**

  $$E(X) = \sum x_i p_i = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = \mathbf{3.5}$$

  The expected number when rolling a fair die is 3.5.

- **2. Calculate the Variance (σ^2):** We use the alternate formula, $\sigma^2 = E(X^2) - [E(X)]^2$.

- First, calculate $E(X^2)$:

$$E(X^2) = \sum x_i^2 p_i = \frac{1+4+9+16+25+36}{6} = \frac{91}{6}$$

- Now, calculate the Variance:

$$\sigma^2 = \frac{91}{6} - \left(\frac{21}{6}\right)^2$$

$$\sigma^2 = \frac{91}{6} - \frac{441}{36} = \frac{91 \times 6}{36} - \frac{441}{36} = \frac{546 - 441}{36} = \frac{105}{36}$$

$$\sigma^2 = \frac{35}{12} \approx \mathbf{2.9167}$$

Example 3: Finding Constant in PMF

Problem: A random variable X takes values $r = 1, 2, 3, \ldots$ with a PMF $P(X = r) = \frac{\lambda^r}{r!}$. Given that the sum of probabilities is valid, find the value of λ.

Key Principle (PMF Condition): The sum of all probabilities must equal 1.

$$\sum_{r=1}^{\infty} P(X = r) = 1$$

1. **Set up the sum:**

$$\sum_{r=1}^{\infty} \frac{\lambda^r}{r!} = 1$$

This series is closely related to the Taylor series expansion of the exponential function, e^x:

$$e^x = \sum_{r=0}^{\infty} \frac{x^r}{r!} = \frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

2. **Compare the PMF sum to the exponential series:**
   The sum $\sum_{r=1}^{\infty} \frac{\lambda^r}{r!}$ is the exponential series e^λ **minus** the first term (where $r = 0$):

$$e^\lambda = \sum_{r=0}^{\infty} \frac{\lambda^r}{r!} = \frac{\lambda^0}{0!} + \sum_{r=1}^{\infty} \frac{\lambda^r}{r!} = 1 + \sum_{r=1}^{\infty} \frac{\lambda^r}{r!}$$

3. **Substitute the PMF sum and solve for λ:**
   Since $\sum_{r=1}^{\infty} \frac{\lambda^r}{r!} = 1$, we substitute this back into the exponential equation:

$$e^\lambda = 1 + 1$$

$$e^\lambda = 2$$

4. **Solve for λ using the natural logarithm:**

$$\lambda = \ln(2)$$

The value of the constant λ is $\ln(\mathbf{2})$. (Note: This type of PMF, $\frac{e^{-\lambda}\lambda^r}{r!}$, is the structure of the **Poisson Distribution**, and here the constant $e^{-\lambda}$ was implied in the problem statement by assuming the full PMF sum equals 1.)

**Example 4: Discrete Probability Distribution

A random variable X has the following probability function values:

$P(X = 0) = 0$
$P(X = 1) = k$
$P(X = 2) = 2k$
$P(X = 3) = 2k$
$P(X = 4) = 3k$
$P(X = 5) = k^2$
$P(X = 6) = 2k^2$
$P(X = 7) = 7k^2 + k$

Find:

1. The value of k.
2. $P(X < 6)$
3. $P(X \geq 6)$
4. $P(3 < X \leq 6)$ (Note: The transcript solves for a specific range similar to this).

This problem requires using the fundamental properties of a **Discrete Probability Mass Function (PMF)** to first solve for the unknown constant k and then calculate various probabilities.

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1. Find the Value of k

The primary condition for any PMF is that the **sum of all probabilities must equal 1** ($\sum P(X = x_i) = 1$).

1. **Set up the equation using the PMF condition:**

$$P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P($$

2. **Substitute the given expressions for the probabilities:**

$$(0) + (k) + (2k) + (2k) + (3k) + (k^2) + (2k^2) + (7k^2 + k) = 1$$

3. **Combine like terms (terms with k and terms with k^2):**
   - k-terms: $k + 2k + 2k + 3k + k = 9k$
   - k^2-terms: $k^2 + 2k^2 + 7k^2 = 10k^2$
4. **Form the quadratic equation:**

$$10k^2 + 9k = 1$$

$$10k^2 + 9k - 1 = 0$$

5. **Solve the quadratic equation using factoring:**

   We look for factors of $10 \times (-1) = -10$ that sum to 9 (which are 10 and -1).

$$10k^2 + 10k - k - 1 = 0$$

$$10k(k+1) - 1(k+1) = 0$$

$$(10k-1)(k+1) = 0$$

6. **Determine possible values for k:**

$$10k - 1 = 0 \implies k = \frac{1}{10}$$

$$k + 1 = 0 \implies k = -1$$

7. **Select the valid value for k:**

   Since all probabilities in a PMF must be non-negative ($P(X = x_i) \geq 0$), we must check the values:

   - If $k = -1$, then $P(X = 1) = k = -1$. Since probability cannot be negative, $\mathbf{k \neq -1}$.
   - If $k = 1/10$, all probabilities are non-negative.

   The valid value of k is $\mathbf{\frac{1}{10}}$ or $\mathbf{0.1}$.

2. Calculate $P(X < 6)$

The probability $P(X < 6)$ is the sum of probabilities for all values of X strictly less than 6: $X = 0, 1, 2, 3, 4, 5$.

$$P(X < 6) = P(0) + P(1) + P(2) + P(3) + P(4) + P(5)$$

Using the simplified sum from Step 3 of finding k: $9k - (P(6) + P(7))$. Alternatively, using the expressions:

$$P(X < 6) = 0 + k + 2k + 2k + 3k + k^2 = 8k + k^2$$

Substitute $k = 1/10$:

$$P(X < 6) = 8\left(\frac{1}{10}\right) + \left(\frac{1}{10}\right)^2 = \frac{8}{10} + \frac{1}{100} = \frac{80}{100} + \frac{1}{100} = \mathbf{\frac{81}{100}}$$

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3. Calculate $P(X \geq 6)$

The probability $P(X \geq 6)$ is the sum of probabilities for $X = 6$ and $X = 7$.

$$P(X \geq 6) = P(6) + P(7)$$

$$P(X \geq 6) = (2k^2) + (7k^2 + k) = 9k^2 + k$$

Alternatively, we can use the complement rule: $P(X \geq 6) = 1 - P(X < 6)$.

$$P(X \geq 6) = 1 - \frac{81}{100} = \mathbf{\frac{19}{100}}$$

Verification (using $9k^2 + k$):

$$9\left(\frac{1}{10}\right)^2 + \frac{1}{10} = \frac{9}{100} + \frac{10}{100} = \mathbf{\frac{19}{100}}$$

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4. Calculate $P(3 < X \leq 6)$

The probability $P(3 < X \leq 6)$ includes values of X strictly greater than 3 and less than or equal to 6: $X = 4, 5, 6$.

$$P(3 < X \leq 6) = P(4) + P(5) + P(6)$$

$$P(3 < X \leq 6) = (3k) + (k^2) + (2k^2) = 3k + 3k^2$$

Substitute $k = 1/10$:

$$P(3 < X \leq 6) = 3\left(\frac{1}{10}\right) + 3\left(\frac{1}{10}\right)^2$$

$$P(3 < X \leq 6) = \frac{3}{10} + \frac{3}{100}$$

$$P(3 < X \leq 6) = \frac{30}{100} + \frac{3}{100} = \mathbf{\frac{33}{100}}$$

**Example 5: Defective Bulbs

5 defective bulbs are accidentally mixed with 20 good ones. It is not possible to just look at a bulb and tell whether or not it is defective. Find the probability distribution of the number of defective bulbs if 4 bulbs are drawn at random from this lot.

This is a problem involving combinations to determine the probabilities, as the order in which the bulbs are drawn does not matter, and the problem fits the structure of a Hypergeometric Distribution.

Here is the step-by-step solution to find the probability distribution.

1. Define the Random Variable and Parameters

- **Total Bulbs (N):** $5 \text{ (defective)} + 20 \text{ (good)} = 25$ bulbs.
- **Total Defective Bulbs (K):** 5.
- **Sample Size Drawn (n):** 4 bulbs.
- **Random Variable (X):** The number of defective bulbs drawn in the sample of 4.

The possible values for the random variable X are the integers from **0 to 4** (since you can draw a minimum of 0 defective bulbs and a maximum of 4 defective bulbs, as you only draw 4 total).

$$X \in \{0, 1, 2, 3, 4\}$$

2. Calculate the Total Possible Cases ($n(S)$)

The total number of ways to draw 4 bulbs from 25 is given by the combination formula NC_n:

$$n(S) = \binom{25}{4} = \frac{25!}{4!(25-4)!} = \frac{25 \times 24 \times 23 \times 22}{4 \times 3 \times 2 \times 1}$$

$$n(S) = 25 \times 23 \times 22 = \mathbf{12,650}$$

3. Calculate the Probability Mass Function (PMF)

The probability of drawing x defective bulbs (from the 5 defective ones) and $(4-x)$ good bulbs (from the 20 good ones) is calculated using the formula:

$$P(X=x) = \frac{\binom{\text{Defective}}{x} \times \binom{\text{Good}}{4-x}}{\binom{\text{Total}}{4}} = \frac{\binom{5}{x} \times \binom{20}{4-x}}{12,650}$$

x	$n(X=x)$ **(Favorable Cases)**	$P(X=x)$ **(Probability)**
0	$\binom{5}{0} \times \binom{20}{4} = 1 \times 4,845 = 4,845$	$\frac{4,845}{12,650} \approx \mathbf{0.3830}$
1	$\binom{5}{1} \times \binom{20}{3} = 5 \times 1,140 = 5,700$	$\frac{5,700}{12,650} \approx \mathbf{0.4506}$
2	$\binom{5}{2} \times \binom{20}{2} = 10 \times 190 = 1,900$	$\frac{1,900}{12,650} \approx \mathbf{0.1502}$
3	$\binom{5}{3} \times \binom{20}{1} = 10 \times 20 = 200$	$\frac{200}{12,650} \approx \mathbf{0.0158}$
4	$\binom{5}{4} \times \binom{20}{0} = 5 \times 1 = 5$	$\frac{5}{12,650} \approx \mathbf{0.0004}$

4. Probability Distribution

The **Probability Distribution** of the number of defective bulbs (X) drawn is:

X **(Number of Defective Bulbs)**	$P(X=x)$
0	$4,845/12,650$
1	$5,700/12,650$
2	$1,900/12,650$
3	$200/12,650$
4	$5/12,650$
Sum	$12,650/12,650 = 1$

That's an excellent move! **Continuous Random Variables** are fundamentally different from discrete ones because they deal with ranges rather than distinct points, which is why we use integration instead of summation.

4. Continuous Random Variable Tools

A **Continuous Random Variable** (X) can take any value within a specific interval. Since there are infinitely many possible values, the probability of X being *exactly* equal to a single point is zero. Instead, we calculate the probability that X falls within a range using the **Probability Density Function (PDF)**.

Probability Density Function (PDF)

The **Probability Density Function ($f(x)$)** is a curve that describes the relative likelihood of the variable taking on a certain value. The probability is represented by the **area under this curve** over a given interval.

- **Conditions for a function to be a PDF:**
  1. **Non-Negativity:** The function value must be non-negative everywhere, as density cannot be negative.

$$f(x) \geq 0 \quad \text{for all } x$$

  2. **Total Area is 1:** The total area under the entire PDF curve must be equal to 1, representing 100% probability for the entire sample space.

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

Probability in an Interval

The probability that the random variable X lies between two values, a and b, is calculated by finding the area under the PDF curve between those two points:

$$P(a < X < b) = \int_{a}^{b} f(x)\, dx$$

Example 6: Finding Constant in PDF (Exponential)

Problem: Find the value of C such that $f(x) = Ce^{-x}$ for $0 \leq x < \infty$ and $f(x) = 0$ otherwise, is a valid PDF.

1. **Apply the Total Area Condition:** Set the integral of the PDF over its domain equal to 1.

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

Since $f(x)$ is non-zero only for $0 \leq x < \infty$:

$$\int_{0}^{\infty} Ce^{-x}\, dx = 1$$

2. **Solve the improper integral:**
   Factor out the constant C:

$$C \int_0^\infty e^{-x}\, dx = 1$$

   Evaluate the integral, noting that $\int e^{-x} dx = -e^{-x}$:

$$C \left[-e^{-x} \right]_0^\infty = 1$$

   Apply the limits of integration:

$$C \left(\lim_{t\to\infty} (-e^{-t}) - (-e^{-0}) \right) = 1$$

3. **Evaluate the limits:**

   - $\lim_{t\to\infty}(-e^{-t}) = -e^{-\infty} = 0$
   - $(-e^{-0}) = -1$

   Substitute these values back:

$$C\,(0 - (-1)) = 1$$

$$C(1) = 1$$

$$\mathbf{C = 1}$$

Example 7: Finding Constant & Probability in PDF (Polynomial)

Problem: A continuous R.V. X has the PDF $f(x) = Cx^2$ for $0 < x < 1$.

1. Find the value of the constant C.

1. **Apply the Total Area Condition:**

$$\int_0^1 f(x)\,dx = 1$$

$$\int_0^1 Cx^2\,dx = 1$$

2. **Solve the integral:**

$$C\left[\frac{x^3}{3}\right]_0^1 = 1$$

3. **Apply the limits:**

$$C\left(\frac{(1)^3}{3} - \frac{(0)^3}{3}\right) = 1$$

$$C\left(\frac{1}{3} - 0\right) = 1$$

$$\frac{C}{3} = 1$$

$$\mathbf{C = 3}$$

2. Find the probability $P\left(\frac{1}{3} < X < \frac{1}{2}\right)$.

Now that we know $C = 3$, the PDF is $f(x) = 3x^2$.

1. **Set up the definite integral for the probability:**

$$P\left(\frac{1}{3} < X < \frac{1}{2}\right) = \int_{1/3}^{1/2} 3x^2\,dx$$

2. **Solve the integral:**

$$P = 3\left[\frac{x^3}{3}\right]_{1/3}^{1/2}$$

$$P = \left[x^3\right]_{1/3}^{1/2}$$

3. **Apply the limits:**

$$P = \left(\frac{1}{2}\right)^3 - \left(\frac{1}{3}\right)^3$$

$$P = \frac{1}{8} - \frac{1}{27}$$

4. **Find a common denominator (LCM of 8 and 27 is 216):**

$$P = \frac{1 \times 27}{216} - \frac{1 \times 8}{216}$$

$$P = \frac{27 - 8}{216} = \mathbf{\frac{19}{216}}$$

The probability that X falls between $1/3$ and $1/2$ is $\frac{19}{216}$.